

A Method for Estimating Temperature of Grasped Object using PPG Sensor

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ABSTRACT

Wearable devices can obtain a variety of information, such as pulse waves, by measuring biometric data. They are applied to many services, including support for the elderly and medical assistance. In the measurement of pulse waves, capillaries dilate and constrict due to changes in skin temperature, which may result in changes in blood flow. Since changes in blood flow are the same as changes in pulse wave, it can be inferred that there is a correlation between skin temperature and pulse sensor readings, which changes depending on the temperature of the object being grasped. In this study, we propose a method for estimating the temperature of a grasped object using a photoplethysmography (PPG) sensor installed in a smartwatch or activity meter. By estimating the grasped object temperature, cold burns and temperature-related sensory disturbances such as unconsciously holding a hot object for a long time in people who have no pain sense due to congenital lack of pain perception or in the elderly who have no pain sense due to a functional decline in the cerebrum caused by dementia can be prevented. In this study, we propose a method to estimate the temperature of an object being touched using a PPG sensor. The proposed method measures the pulse wave from the PPG sensor and estimates the temperature from the rate of change of the difference between the peak of the pulse wave after touching the object and the pulse wave under normal conditions. The results of the temperature estimation experiment for liquids on four subjects showed that the proposed method could estimate the temperature with a 78.1% correct rate, and for solids on two subjects the rate was 75%.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

KEYWORDS

Object temperature estimation, PPG sensor, pulse wave

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1 INTRODUCTION

Wearable devices such as smartwatches and activity meters have been developed and are becoming widely available. Data on the user wearing the device can be acquired without affecting other tasks or to view information while the user is working. These devices are also used in factories and other business environments, as worker management and productivity can be improved. Specifically, there have been proposed services such as health management systems[5] that automatically extract lifestyle patterns using data acquired from wearable devices and warn against lack of exercise and overwork. General wearable devices do not have the ability to measure the temperature of objects that the user is in contact with, although some are equipped with a body temperature sensor or an outside temperature sensor. The information obtained from wearable devices, especially pulse waves, has a feature that tends to change depending on the surrounding environment. Therefore, this study proposes a method to estimate the temperature of an object in contact with the user using a PPG sensor in a smartwatch or activity meter. The proposed method uses the rate of change of the difference between the peak values of the pulse wave before and after the user's contact with the object. The temperature is estimated by comparing it with previously prepared training data. Since the proposed method can estimate the temperature of the contacted object from the pulse wave, it has the advantage of preventing dangerous conditions caused by temperature, such as cold burns and frostbite in people with temperature-related sensory disorders.

2 RELATED WORK

There have been proposed methods that estimate temperature. Hamatani et al.[3] propose a method for estimating deep body temperature by simulation. Both papers proposed a method for estimating body core temperature based on a two-node human body thermal model using wearable sensors. In addition, an application[9] was developed to estimate the user's body temperature based on the body heat model and the vital sensors in the smartwatch. Funahashi et al.[2] discussed a model for the validity of transmitted power density (TPD) at the skin as a new indicator for estimating steady skin temperature rise above the transition frequency. The results showed that the model could be used to estimate elevated skin temperature with high accuracy. Angela et al.[7] evaluated whether rumen temperature could be used as a surrogate indicator of core body temperature in fattening cattle by determining the relationship between the rectal temperature and rumen temperature.

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Their study showed a moderate to strong relationship between the rectal temperature and rumen temperature using air temperature, relative humidity, wind speed, wind direction, solar radiation, and black bulb temperature as parameters.

Pulse wave data have been used in many works. Havriushenko et al.[1] proposed a method for estimating respiratory rate from pulse wave data by using neural networks. Kameyama et al.[6] developed an algorithm that measures pulse waves during sleep with a wristwatch device and estimates two sleep states. They also apply arousal stimuli to each sleep state and investigate the effects of sleep state and sleep state switching timing on sleep quality. Kajiware et al.[10] focused on the fact that many logistics companies adopt manual order picking systems and that emotions and engagements affect work efficiency and human errors, and proposed a method to predict emotions and engagements during high-intensity work based on behaviors and pulse waves acquired by wearable devices. Han et al.[4] developed an algorithm to detect premature atrial contraction and premature ventricular contraction using PPG data acquired from a smartwatch. These studies have used pulse wave data obtained by a PPG sensor for emotion estimation and sleep assistance. However, to the best of our knowledge, no study has been conducted to estimate temperature using pulse wave data.

3 PROPOSED METHOD

This section explains the preliminary experiment to see the relationship between pulse wave and temperature, and the process of the proposed method.

3.1 Pulse Wave and Temperature Relationship

A preliminary experiment was conducted to confirm the relationship between PPG sensor values and the temperature of objects in contact with the hands. Two subjects (A and B, both male, average age 24) joined the experiment. The subjects immersed their hand in the water tank up to the wrists for one minute. After one minute, the hands were thoroughly wiped dry, and pulse wave data were measured for 30 seconds. The water temperature was randomly set in the range from 0 to 50°C. Pulse wave measurements were taken ten times for each subject.

Figure 1 shows the results of a linear regression between the average of the peak pulse wave data collected from the two subjects and the temperature of the water in the tank in which their hands were immersed. This figure shows that the correlation coefficient between the pulse wave peak value and the temperature of the water touched by the two subjects is greater than 0.9, indicating a strong positive correlation between the PPG sensor value and the temperature of the object the hand is touching. Based on this result, this paper proposes a method that estimates the temperature of the object in contact with the hand from the pulse wave data.

3.2 Overview of the Proposed Method

Figure 2 shows the flow of the proposed method. The proposed method consists of four processes: pulse wave sensing, peak detection, peak-to-peak value calculation, and temperature estimation.

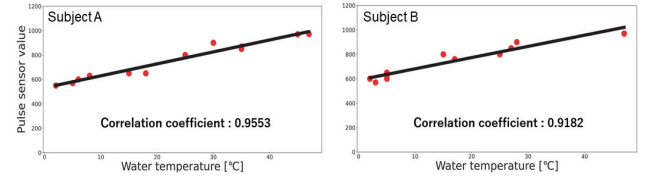


Figure 1: Regression line between the peak value of pulse wave data and water temperature.

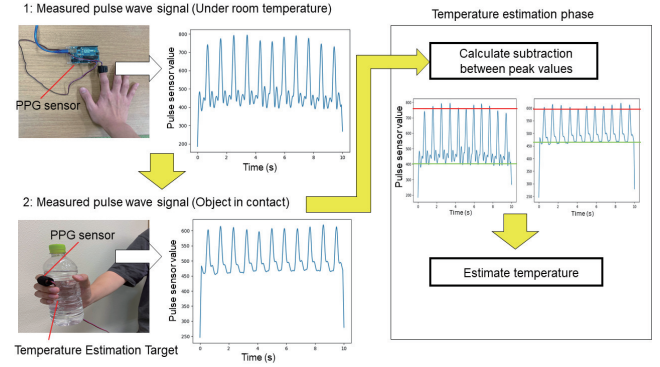


Figure 2: Overview of the proposed method.

3.3 Pulse Wave Sensing

The proposed method assumes that a wearable device such as a smart ring equipped with a PPG sensor is attached to the user's finger. The photoelectric method uses an LED to irradiate light with a green wavelength near 550 nm onto the skin surface, and a photo-diode on the skin surface receives the irradiated light reflected by body tissues. Since hemoglobin absorbs incident light, the reflected light decreases at the moment of pulse wave arrival due to the increase in blood volume, and the pulse rate is measured. The proposed method estimates temperature using pulse wave data obtained at a certain interval. Let $p(t)$ be the pulse wave measurement at time t obtained by the PPG sensor.

3.4 Peak Detection

Peaks that appear in the acquired pulse wave are detected. Sensor data often contains noise caused by the analog-to-digital converter (ADC). Therefore, a moving average filter with a length window of 11 samples (≈ 0.22 seconds) is used to obtain the pulse wave $p'(t) = \sum_{i=t-5}^{t+5} p(i)$. Peak detection is performed on the filtered pulse wave $p'(t)$. Peak detection is performed using `argrelmax` and `argrelmin` functions in the Python SciPy[8] package. `argrelmax` is a function to detect maxima and `argrelmin` is a function to detect minima. Using these two functions, the maxima and minima in $p'(t) (0 \leq t \leq T)$ are detected exhaustively. Note that T is the length of the acquired pulse wave data after the moving average filter. Temperature estimation is performed using pulse wave data $p'(t)$ after moving average filtering cut out in a sliding window of $T = 30$ seconds. The detected maxima and minima in the sliding window are stored in arrays named V_{max} and V_{min} , respectively.

3.5 Peak-to-peak Value Calculation

The difference between the mean of the maxima and the mean of the minima in $p'(t)$ is calculated as peak-to-peak value v_{pp} . When the number of maxima in V_{max} is n_{max} , the mean maxima $m_{max} = \frac{1}{n_{max}} \sum_{t=0}^{n_{max}} V_{max}[t]$ is calculated. When the number of minima in V_{min} is n_{min} , the mean minima $m_{min} = \frac{1}{n_{min}} \sum_{t=0}^{n_{min}} V_{min}[t]$ is calculated. At last, peak-to-peak value v_{pp} is calculated with $d = m_{max} - m_{min}$. The d calculated at room temperature and after object contact is denoted d_{normal} and d_{test} , respectively.

3.6 Temperature Estimation

The proposed method estimates temperature using the ratio r between d_{normal}, v_{pp} at room temperature, and d_{test}, v_{pp} after contact with an object. However, the difference between the peak values varies depending on the user. Therefore, we collect pulse wave data for each temperature range in advance and calculate r to use as training data. The r can be expressed as $r = \frac{d_{test}}{d_{normal}}$, where r is calculated by the proposed method estimates the temperature in the range of 10°C separations.

4 EVALUATION EXPERIMENT

An experiment was conducted to evaluate the performance of estimating temperature of objects in contact with the hand.

4.1 Setup

Four subjects (A–D, all male, average 23.5 years) joined the experiment. The subjects were asked to submerge their right hands up to the wrists in a water tank filled with water maintained at a specified temperature for one minute. After one minute of immersion, the subjects put their hands out from the tank, wiped them thoroughly dry, and attached a PPG sensor to the index fingertip of their right hand. The PPG values were recorded at 50 Hz.

We further conducted an experiment in which two subjects (A and B, all male, average 25.0 years) were asked to hold a PET bottle filled with water set at four different temperatures in their hands. The subjects grasped the PET bottle for one minute, and then the PPG sensor was attached to the base of the index finger of their right hand. The reason for attaching the sensor to the base of the finger is that it is most likely to come into contact with the object when the subject grasps it. The PPG values were recorded at 50 Hz.

For both experiments, water tank and plastic bottles filled with water, four water temperatures were selected: 0–10°C, 10–20°C, 20–30°C, and 40–50°C. The 30–40°C is not included in the experiment because it is close to the surface temperature of the human body and is considered to be unaffected by water temperature. The room temperature was always kept at around 25°C in order not to affect the pulse wave height. To eliminate the influence of ambient factors such as light, the pulse wave was measured with a finger covered with black Velcro tape.

4.2 Data Collection

First, subjects were asked to wear a PPG sensor to collect data for the study. Five 30-second data were collected at each temperature range. As a result, 4 subjects $\times$ 30 seconds $\times$ 5 times $\times$ 4 types = 40 minutes of data were collected when dipping their hands into the

Table 1: Temperature determination model using the maxima and minima (peak-to-peak) of pulse wave for water tank.

Subject	0–10°C	10–20°C	20–30°C	40–50°C
A	<0.412	0.412–0.889	0.889–0.947	1.000<
B	<0.565	0.565–0.714	0.714–0.938	1.000<
C	<0.561	0.561–0.875	0.875–0.966	1.000<
D	<0.230	0.230–0.468	0.468–0.938	1.000<

Table 2: Temperature determination model using the maxima (peak) of pulse wave for water tank.

Subject	0–10°C	10–20°C	20–30°C	40–50°C
A	<0.753	0.753–0.876	0.876–0.980	1.000<
B	<0.656	0.656–0.810	0.810–0.969	1.000<
C	<0.675	0.675–0.872	0.872–0.978	1.000<
D	<0.214	0.214–0.659	0.659–0.977	1.000<

Table 3: Temperature determination model using the maxima and minima of the pulse wave for PET bottle.

Subject	0–10°C	10–20°C	20–30°C	40–50°C
A	<0.525	0.525–0.675	0.675–0.900	1.000<
B	<0.692	0.692–0.843	0.843–0.980	1.000<

liquid, and 2 subjects $\times$ 30 seconds $\times$ 5 times $\times$ 4 types = 20 minutes when grasping a PET bottle. From the collected data, the ratio R of the difference between the maximum and minimum pulse wave values was calculated by rounding to four decimal places.

Using the previously collected data, a model for determining the temperature from the values at each temperature range for each subject is shown in Table 1. For comparison, the model for determining a temperature from the ratio of change in only the maximum value of pulse wave data, r_{max} , is shown in Table 2. Furthermore, the temperature prediction model for grasping a PET bottle is shown in Table 3.

Next, data were collected for the evaluation experiment. As with the training data, a PPG sensor was attached and data was collected twice for 30 seconds at each temperature range. As a result, data was collected for 4 subjects $\times$ 30 seconds $\times$ 4 types = 8 minutes in the case of putting hands in water, and for 2 subjects $\times$ 30 seconds $\times$ 4 types = 4 minutes in the case of grasping a plastic bottle. From the collected data, r , the ratio of the difference between peak values, was calculated for each of the cases of dipping the hand in the water and grasping the PET bottle. r_{max} , the rate of change of only the maxima, was also calculated in the case of dipping the hand in the water. In the proposed method, four decimal places were rounded off. In this experiment, we use the models shown in Table 1, Table 2 and Table 3 to estimate the temperature for the test data. These tables show the r -values calculated from experiments conducted on four subjects in each situation.

4.3 Results and Discussion

Table 4 shows the results of the temperature estimation for the dipping of the hand in water using the proposed method. The

Table 4: Results of temperature estimation for water tank.

Subject	#	Peak-to-peak				Peak			
		r	Input	Output		r_{max}	Input	Output	
A	1	0.138	0–10°C	0–10°C	✓	0.587	0–10°C	0–10°C	✓
	2	0.278	0–10°C	0–10°C	✓	0.632	0–10°C	0–10°C	✓
	3	0.455	10–20°C	10–20°C	✓	0.867	10–20°C	10–20°C	✓
	4	0.700	10–20°C	10–20°C	✓	0.713	10–20°C	0–10°C	
	5	0.722	20–30°C	10–20°C		0.762	20–30°C	10–20°C	
	6	0.763	20–30°C	10–20°C		0.723	20–30°C	10–20°C	
	7	1.200	40–50°C	40–50°C	✓	1.213	40–50°C	40–50°C	✓
	8	1.370	40–50°C	40–50°C	✓	1.157	40–50°C	40–50°C	✓
B	1	0.208	0–10°C	0–10°C	✓	0.632	0–10°C	0–10°C	✓
	2	0.405	0–10°C	0–10°C	✓	0.773	0–10°C	10–20°C	
	3	0.506	10–20°C	0–10°C		0.868	10–20°C	20–30°C	
	4	0.605	10–20°C	10–20°C	✓	0.950	10–20°C	20–30°C	
	5	0.373	20–30°C	0–10°C		0.908	20–30°C	20–30°C	✓
	6	0.883	20–30°C	20–30°C	✓	0.918	20–30°C	20–30°C	✓
	7	1.030	40–50°C	40–50°C	✓	1.210	40–50°C	40–50°C	✓
	8	1.145	40–50°C	40–50°C	✓	1.215	40–50°C	40–50°C	✓
C	1	0.083	0–10°C	0–10°C	✓	0.742	0–10°C	10–20°C	
	2	0.460	0–10°C	0–10°C	✓	0.615	0–10°C	0–10°C	✓
	3	0.600	10–20°C	10–20°C	✓	0.456	10–20°C	0–10°C	
	4	0.823	10–20°C	10–20°C	✓	0.886	10–20°C	20–30°C	
	5	0.842	20–30°C	10–20°C		0.617	20–30°C	10–20°C	
	6	0.926	20–30°C	20–30°C	✓	0.884	20–30°C	20–30°C	✓
	7	1.011	40–50°C	40–50°C	✓	1.130	40–50°C	40–50°C	✓
	8	1.176	40–50°C	40–50°C	✓	2.660	40–50°C	40–50°C	✓
D	1	0.214	0–10°C	0–10°C	✓	0.880	0–10°C	20–30°C	
	2	0.230	0–10°C	0–10°C	✓	0.884	0–10°C	20–30°C	
	3	0.118	10–20°C	0–10°C		0.733	10–20°C	20–30°C	
	4	0.356	10–20°C	10–20°C	✓	0.866	10–20°C	20–30°C	
	5	0.210	20–30°C	0–10°C		0.936	20–30°C	20–30°C	✓
	6	0.938	20–30°C	20–30°C	✓	0.757	20–30°C	20–30°C	✓
	7	1.200	40–50°C	40–50°C	✓	1.018	40–50°C	40–50°C	✓
	8	1.802	40–50°C	40–50°C	✓	1.140	40–50°C	40–50°C	✓

results show that 25 out of 32 trials were successful in estimating the temperature using the model of the difference r between the peak values. The percentage of correct answers was 78.1 percent. In particular, temperature estimation was successful for all subjects when the actual temperature range was less than 10°C and when the actual temperature range was 40–50°C. This may be because vasoconstriction and dilation were more pronounced at cold and hot temperatures. In addition, incorrect estimation results were output in some trials in the temperature ranges of 10–20°C and 20–30°C. This is believed to be because the difference between body temperature and the temperature of the contacting object is small and the contraction of blood vessels is also small, making it difficult for the rate of change to differ.

On the other hand, using a model with a maximum value of r_{max} , the temperature was successfully estimated in 18 of the 32 trials. The correct estimation rate was 56.3%, which is a low result. The reason for this is that not only the maxima but also the minima of the pulse waveform change with changes in temperature, making it difficult to estimate the temperature only by the maxima. The

average change of the maxima was small, while the average change of the minima was large. This can be considered to be due to the fact that the blood flow that should have been flowing at the time of pulse wave arrival is suppressed by vasoconstriction, resulting in an increase in blood flow outside of the heartbeat. From the above results, the model with the difference r between peak values has a higher accuracy rate than the model with the maximum value R_{max} , suggesting that the model with r in the proposed method is useful.

Next, Table 5 shows the results of grasping a PET bottle. Based on the results, the temperature of the grasped object was estimated using a model of the difference between the peak values, which was successful in 12 out of 16 trials. The correct estimation rate was 75%. As in the experiment with the hand dipped in water, the estimation results at 10–20°C and 20–30°C were not good. This is thought to be due to the fact that the pulse wave is less likely to change when the temperature of the grasped object is close to the body temperature.

Table 5: Results of temperature estimation for PET bottle.

Subject	#	r	Peak-to-peak		
			Input	Output	
A	1	0.163	0–10°C	0–10°C	✓
	2	0.400	0–10°C	0–10°C	✓
	3	0.254	10–20°C	0–10°C	
	4	0.537	10–20°C	10–20°C	✓
	5	0.364	20–30°C	0–10°C	
	6	0.825	20–30°C	20–30°C	✓
	7	1.204	40–50°C	40–50°C	✓
	8	2.833	40–50°C	40–50°C	✓
B	1	0.649	0–10°C	0–10°C	✓
	2	0.474	0–10°C	0–10°C	✓
	3	0.757	10–20°C	10–20°C	✓
	4	0.911	10–20°C	20–30°C	
	5	0.662	20–30°C	0–10°C	
	6	0.923	20–30°C	20–30°C	✓
	7	1.009	40–50°C	40–50°C	✓
	8	1.237	40–50°C	40–50°C	✓

PPG sensor values easily change depending on the environment in which the user is located, the user's physical condition on that day, and the state of contact with the sensor. Therefore, it is necessary to estimate the temperature by taking these factors into account. Therefore, it is necessary to evaluate the model in different environments for the training data and the test data, such as after exercise. In addition, since the number of subjects was only four, it is considered necessary to conduct experiments using a larger number of subjects with a wide age range. In addition, since wrist-worn wearable devices such as smart rings and smart watches have been available in recent years, we believe it is necessary to conduct experiments in which pulse waves are measured at the base of the fingers or on the wrist, rather than at the fingertips.

5 CONCLUSION

In this paper, we proposed a method for estimating the temperature of an object in contact with the skin using changes in the pulse wave. The proposed method measures the pulse wave from a PPG sensor and estimates the temperature based on the percentage of change in the difference between the peak values of the pulse wave after touching an object and the pulse wave during normal conditions. The experiment with four subjects using water tank showed 78.1% accuracy, and the experiment with two subjects using PET bottle showed 75% accuracy.

We will improve the proposed method so that the estimated temperature range can be finer and evaluate the system at different sensor positions. In addition, since PPG sensor values can easily change depending on the sensor position and user's physical condition, we plan to investigate a correction method for temperature estimation that takes these factors into account.

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